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By Amit Katiyar (MCA-JNU)



MATHEMATICS

- Area of the parallelogram formed by the lines $y = 4x$, $y = 4x + 1$, $x + y = 0$ and $x + y = 1$ is
(a) $\frac{1}{5}$ (b) $\frac{2}{5}$ (c) 5 (d) 10
- The Function $f(x) = \begin{cases} (1+2x)^{\frac{1}{x}}, & x \neq 0 \\ e^2, & x = 0 \end{cases}$ is:
(a) differentiable at $x = 0$ (b) continuous at $x = 0$
(c) discontinuous at $x = 0$ (d) not differentiable at $x = 0$
- If $x^m y^n = (x+y)^{m+n}$, then $\frac{dy}{dx}$ is
(a) $\frac{x+y}{xy}$ (b) xy (c) $\frac{x}{y}$ (d) $\frac{y}{x}$
- A straight line through the point (4, 5) is such that its intercept between the axes is bisected at A, then its equation is
(a) $3x + 4y = 20$ (b) $3x - 4y + 7 = 0$
(c) $5x - 4y = 40$ (d) $5x + 4y = 40$
- If $a = \lambda \hat{i} + \hat{j} - 2\hat{k}$, $b = \hat{i} + \lambda \hat{j} - 2\hat{k}$, $c = \hat{i} + \hat{j} + \hat{k}$ and $[a \ b \ c] = 7$, then the values of λ are
(a) 2, -6 (b) 4, -2 (c) 4, 2 (d) -4, 2
- The value of $3^{3-\log_3 5}$ is
(a) $\frac{5}{27}$ (b) $\frac{27}{5}$ (c) $\frac{9}{5}$ (d) $\frac{5}{9}$
- If $a_1, a_2, \dots, a_n$ be arithmetic progression with common difference, d , then the sum $\sin d (\operatorname{cosec} a_1 \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \operatorname{cosec} a_3 + \dots + \operatorname{cosec} a_{n-1} \operatorname{cosec} a_n)$ is equal to
(a) $\cot a_1 - \cot a_n$ (b) $\sin a_1 - \sin a_n$
(c) $\operatorname{cosec} a_1 - \operatorname{cosec} a_n$ (d) $a_1 - a_n$
- The value of $\cot(\operatorname{cosec}^{-1} \frac{5}{3} + \tan^{-1} \frac{2}{3})$ is
(a) $\frac{6}{17}$ (b) $\frac{3}{17}$ (c) $\frac{4}{17}$
(d) $\frac{5}{17}$
- If $a_1, a_2, \dots, a_n$ are any real number and n is any positive integer, then
(a) $n \sum_{i=1}^n a_i^2 < (\sum_{i=1}^n a_i)^2$ (b) $n \sum_{i=1}^n a_i^2 \geq (\sum_{i=1}^n a_i)^2$
(c) $\sum_{i=1}^n a_i^2 \geq (\sum_{i=1}^n a_i)^2$ (d) None of these
- If $\cos^{-1} \frac{x}{2} + \cos^{-1} \frac{y}{3} = \emptyset$, then $9x^2 - 12xy \cos \emptyset + 4y^2$ is equal to
(a) $36 \sin^2 \emptyset$ (b) $36 \cos^2 \emptyset$
(c) $36 \cos^2 \emptyset$ (d) 36
- If the root of the quadratic equation $x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$, respectively, then the value of $2 + p - q$ is
(a) 3 (b) 0 (c) 1 (d) 2
- If $\operatorname{cosec} \theta - \cot \theta = 2$. Find $\operatorname{cosec} \theta$ is
(a) $\frac{5}{3}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$ (d) $\frac{5}{4}$
- Solution of the equation $\tan^{-1} \sqrt{x^2 + x} + \sin^{-1} \sqrt{x^2 + x + 1} = \frac{\pi}{2}$ are
(a) 0, 1 (b) 1, -1 (c) 0, -1 (d) 0, -2
- If $a < b$, then $\int_a^b (|x-a| + |x-b|) dx$, equal to
(a) $\frac{(b-a)^2}{2}$ (b) $\frac{(b^2-a^2)}{2}$
(c) $\frac{(b^3-a^3)}{2}$ (d) $(b-a)^2$

- Let $a = 2\hat{i} + 2\hat{j} + \hat{k}$ and b be another vector such that $a \cdot b = 14$ and $a \times b = 3\hat{i} + \hat{j} - 8\hat{k}$ the vector $b =$
(a) $5\hat{i} + \hat{j} + 2\hat{k}$ (b) $5\hat{i} - \hat{j} - 2\hat{k}$
(c) $5\hat{i} + \hat{j} - 2\hat{k}$ (d) $3\hat{i} + \hat{j} + 4\hat{k}$
- Let a be the distance between lines $-2x + y = 2$ and $2x - y = 2$, and b be the distance between the lines $4x - 3y = 5$ and $6y - 8x = 1$, then
(a) $40b = 11\sqrt{5}a$ (b) $40\sqrt{2}a = 11b$
(c) $11\sqrt{2}b = 40a$ (d) $11\sqrt{2}a = 40b$
- The area enclosed within the curve $|x| + |y| = 2$ is
(a) 16 sq units (b) 24 sq units
(c) 32 sq units (d) 8 sq units
- In a harmonic progression, p th term is q and q th term is p . Then, pq th term is
(a) 0 (b) 1 (c) pq (d) $pq(p+q)$
- Suppose that the temperature at a point (x, y) on a metal plate is $T(x, y) = 4x^2 - 4xy + y^2$. An ant, walking on the plate, traverses a circle of radius 5 centred at the origin. What is the highest temperature encountered by the ant?
(a) 125 (b) 120 (c) 0 (d) 25
- If $(\frac{x}{a})^2 + (\frac{y}{b})^2 = 1$, $(a > b)$ and $x^2 - y^2 = c^2$ cut at right angles, then
(a) $a^2 + b^2 = 2c^2$ (b) $b^2 - a^2 = 2c^2$
(c) $a^2 - b^2 = 2c^2$ (d) $a^2 - b^2 = c^2$
- If $(\hat{a} \times \hat{b}) \times \hat{c} = \hat{a} \times (\hat{b} \times \hat{c})$, then
(a) a and c are collinear (b) a and c are perpendicular
(c) a and c are collinear (d) a and c are perpendicular
- Angle of elevation of the top of the tower from 3 points (collinear) A, B and C on road leading to the foot of tower are $30^\circ, 45^\circ$ and 60° , respectively. The ratio of AB and BC is
(a) $\sqrt{3}:1$ (b) $\sqrt{3}:2$ (c) 1:2 (d) 2: $\sqrt{3}$
- Let a, b and c be distinct non-negative number. If the vector $a\hat{i} + a\hat{j} + c\hat{k}, \hat{i} + \hat{k}$ and $\hat{c}\hat{i} + \hat{c}\hat{j} + b\hat{k}$ lie in a plane then c is equal to
(a) the arithmetic means of a and b
(b) the geometric mean of a and b
(c) the harmonic mean of a and b
(d) equal to zero
- There are two sets A and B with $|A| = m$ and $|B| = n$. If $|P(A)| - |P(B)| = 112$, then choose the wrong option (where $|A|$ denotes the cardinality of A, and $P(A)$ denotes the power set of A)
(a) $m + n = 11$ (b) $2n - m = 1$
(c) $2m - n = 1$ (d) $3n - m = 5$
- The mean of 25 observations was found to be 38. It was later discovered that 23 and 38 were misread as 25 and 36, then the mean is
(a) 32 (b) 36
(c) 38 (d) 42
- There are two circles in xy -plane whose equations are $x^2 + y^2 - 2y = 0$ and $x^2 + y^2 - 2y - 3 = 0$. A point (x, y) is chosen at random inside the larger circle. then, the probability that the point has been taken from smaller circle is
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$



27. The value of $\int \frac{(x^2-1)dx}{x^3\sqrt{2x^4-2x^2+1}}$ is

- (a) $2\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + c$ (b) $2\sqrt{2 + \frac{2}{x^2} + \frac{1}{x^4}} + c$
 (c) $\frac{1}{2}\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + c$ (d) None of these

28. If $0 < P(A) < 1$ and $0 < P(B) < 1$ and $P(A \cap B) = P(A) \cdot P(B)$, then

- (a) $P(B|A) = P(B) - P(A)$ (b) $P(A^c - B^c) = P(A^c) - P(B^c)$
 (c) $P(A \cup B)^c = P(A^c)P(B^c)$ (d) $P(A|B) = P(A) - P(B)$

29. Which of the following is not true?

- (a) $\lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$ (b) $\lim_{x \rightarrow 0^+} \frac{1}{xe^x} = 0$
 (c) $\lim_{x \rightarrow 0^+} \frac{\sin x}{1+2x} = 0$ (d) $\lim_{x \rightarrow 0^+} \frac{\cos x}{1+2x} = 0$

30. A four digit number is formed using the digit 1,2,3,4 and 5 without repetition. The probability that it is divisible by 3 is

- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{1}{5}$ (d) $\frac{1}{6}$

31. The correct expression for $\cos^{-1}(-x)$ is

- (a) $\frac{\pi}{2} - \cos^{-1} x$ (b) $\pi - \cos^{-1} x$
 (c) $\pi + \cos^{-1} x$ (d) $\frac{\pi}{2} + \cos^{-1} x$

32. If α and β are the roots of $x^2 - x - 1 = 0$, and $A_n = \alpha^n + \beta^n$, then arithmetic mean of A_{n-1} and A_n is

- (a) $2A_{n-1}$ (b) $\frac{1}{2}A_{n+1}$
 (c) $2A_n - 2$ (d) None of these

33. If $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2+x & 1 \\ 1 & 1 & 2+y \end{vmatrix}$, $x \neq 0, y \neq 0$, then D is

- (a) divisible by x and y
 (b) divisible by x but not by y
 (c) divisible by (x+1) and (y+1)
 (d) divisible by (1+x) but not (1+y)

34. In ΔABC , if the tangent of half the difference of two angles is equal to one third of the tangent of the sum of the angles, then the ratio of the sides opposite to the angle is

- (a) 2:1 (b) 1:2 (c) 3:1 (d) 1:1

35. If the volume of a parallelopiped whose adjacent edges are $a = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $b = \hat{i} + \alpha\hat{j} + 2\hat{k}$, $c = \hat{i} + 2\hat{j} + \alpha\hat{k}$ is 15, then α is equal to

- (a) 1 (b) $\frac{5}{2}$ (c) $\frac{9}{2}$ (d) 0

36. The first three moments of a distribution about 2 are 1,16,40, respectively. Then, mean and variance of the distribution are

- (a) (2,16) (b) (2,15) (c) (3,15)
 (d) (1,16)

37. Inverse of the function $f(x) = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$ is

- (a) $\log_{10}(2-x)$ (b) $\frac{1}{2}\log_{10}\left(\frac{1+x}{1-x}\right)$
 (c) $\frac{1}{2}\log_{10}(2x-1)$ (d) $\frac{1}{4}\log_{10}\left(\frac{2x}{2-x}\right)$

38. Coordinate of focus of the parabola $4y^2 + 12x - 20y + 67 = 0$ is

- (a) $\left(-\frac{5}{4}, \frac{17}{4}\right)$ (b) $\left(-\frac{17}{2}, \frac{5}{4}\right)$
 (c) $\left(-\frac{17}{4}, \frac{5}{2}\right)$ (d) $\left(-\frac{5}{2}, \frac{17}{4}\right)$

39. Which term of the series $\frac{\sqrt{5}}{3}, \frac{\sqrt{5}}{4}, \frac{1}{\sqrt{5}}, \dots$ is $\frac{\sqrt{5}}{13}$?

- (a) 12 (b) 11 (c) 10
 (d) 9

40. A particle is at rest at the origin. It moves along the X-axis with an acceleration $x - x^2$, where x is the distance of the particle at time t. The particle next comes to rest after it has covered a distance.

- (a) 1 (b) $\frac{1}{2}$ (c) $\frac{3}{2}$ (d) 2

41. The function $f(x) = \log(x + \sqrt{x^2 + 1})$ is

- (a) An even function
 (b) an odd function
 (c) a periodic function
 (d) neither an even nor an odd function

42. The domain of the function $f(x) = \frac{\cos^{-1} x}{|x|}$ is

- (a) $[-1, 0) \cup \{1\}$ (b) $[-1, 1]$
 (c) $[-1, 1]$ (d) None of these

43. A survey is done among population of 200 people who like either tea or coffee. It is found that 60% of the population like tea and 72% of the population like coffee. Let x be the number of people who like both tea and coffee. Let $m \leq x \leq n$, then choose then correct option.

- (a) $n-m=56$ (b) $n-m=28$
 (c) $n-m=32$ (d) $n+m=92$

44. The solution of the equation $4\cos^2 x + 6\sin^2 x = 5$ are

- (a) $x = n\pi \pm \frac{\pi}{4}$ (b) $x = n\pi \pm \frac{\pi}{3}$
 (c) $x = n\pi \pm \frac{\pi}{2}$ (d) $x = n\pi \pm \frac{2\pi}{3}$

45. $f(x) = x + |x|$ is continuous for

- (a) $x \in (-\infty, \infty)$ (b) $x \in (-\infty, \infty) - \{0\}$
 (c) only $x > 0$ (d) no value of x

46. The 10th and 50th percentiles of the observation 32, 49, 23, 29 and 118, respectively are

- (a) 21, 32 (b) 23, 32
 (c) 23, 33 (d) 22, 31

47. For $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$

$$\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1}[(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$$

Then one of the value of a is (a) 5 (b) 8

- (c) $-\frac{15}{2}$ (d) $-\frac{17}{2}$

48. The eccentricity of an ellipse, with its centre at the origin is $1/3$.

If one of the directrix is $x=9$, then the equation of ellipse is

- (a) $9x^2 + 8y^2 = 72$ (b) $8x^2 + 9y^2 = 72$
 (c) $8x^2 + 7y^2 = 56$ (d) $7x^2 + 8y^2 = 56$

49. If the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{b^2} = 1$ and hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = 1$ are coincide, then the value of b^2 is

- (a) 25 (b) 16 (c) 64 (d) 49

50. If the angle of elevation of the top of a hill from each of the vertices A, B and C of a horizontal is a, then the height of the hill is

- (a) $\frac{1}{2}b \tan \alpha \sec B$ (b) $\frac{1}{2}b \tan \alpha \operatorname{cosec} A$
 (c) $\frac{1}{2}c \tan \alpha \sin C$ (d) $\frac{1}{2}a \tan \alpha \operatorname{cosec} A$



LOGICAL REASONING

51. Select the pair of words, which are related in the same way as the capitalized word are related to each other.

Frugal : Extravagant

- (a) Predecessor : Precursor (b) Criticise : Advocate
(c) Teacher : Philanthropist (d) Hermit : Philosopher

52. Fill in blank : HEC , JGE , LIG , NKI , ...

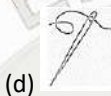
- (a) PMK (b) HGF (c) KMP (d) ONM

53. Replace the question mark with an appropriate image to complete

The analogous pair.



(c)



54. Deepak, Rahul, Manoj and Vinod are brothers. Who is the heaviest?

I. Rahul is heavier than Deepak and Vinod, but lighter than Manoj.

II. Deepak is lighter than Rahul and Manoj.

But heavier than Vinod

- (a) Data in both the statements together are not sufficient.
(b) Statement II alone is sufficient, but Statement I alone is not sufficient.
(c) Either I or II sufficient.
(d) Statement I alone is sufficient, but Statement II alone is not sufficient.

55. Select the related word from the given alternatives. MELT: LIQUID :: FREEZE : ?

- (a) PUSH (b) CONDENSE
(c) ICE (d) SOLID

56. Identify the fifth number in the series: 122, 144, 166, 188, ?

- (a) 210 (b) 310 (c) 345
(d) 234

57. Today is Wednesday. What would be the day after 61 days?

- (a) Sunday (b) Monday (c) Tuesday
(d) Saturday

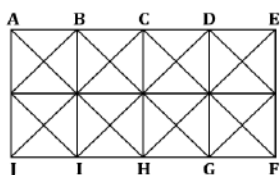
58. Fill in the blank : JAK, KBL LCM, MDN,

- (a) NEO (b) OEP (c) MEN (d) PEQ

59. DNN, FQQ, HTT, , LZZ

- (a) JWW (b) IWW (c) JVV (d) JXX

60. In the following figure. Find the total number of squares.



- (a) 24 (b) 20
(c) 36 (d) 18

61. Find the synonym that is most nearly similar in meaning to the word: DEBACLE

- (a) Catastrophe (b) Dandy
(c) Opulence (d) Corker

62. Looking at the portrait of a man, Manu said, "Her mother is the wife of my father's son and I have no brother and sister." Whose portrait was Manu looking at ?

- (a) His son (b) His daughter
(c) His nephew (d) His father

63. Identify the sixth number in the series: 6, 11, 21, 36, 56, ?

- (a) 82 (b) 21 (c) 81
(d) 52

64. U, V, W, X and T are sitting on a bench. T is sitting next to U, V is sitting next to W, W is not sitting with X who is on the left end of the bench. V is in the second position from the right. T is to the right of U and X. T and V are sitting together. In which position T is sitting?

- (a) Between U and V (b) Between X and W
(c) Between V and X (d) Between U and W

65. If in a certain language, KOLKOTA is coded as LPMLPUB, how is MUMBAI coded in that code?

- (a) NVNBCJ (b) OVNBH (c) NUNBCH (d) NVNCBJ

66. Select the pair of words, which are related in the same way as the capitalized words are related to each other.

Butterfly : freedom

- (a) Frog : Water (b) Self-reliant : Buoyant
(c) Horse : Speed (d) Chicken : Rooster

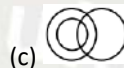
67. Which of the following diagrams correctly represents lions, elephants and animals?



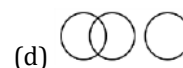
(a)



(b)



(c)



(d)

68. Choose the word opposite in meaning to the given word: MITIGATE

- (a) Abate (b) Tranquelize
(c) Intensify (d) Alleviate

69. Which of the following is to odd one from the given alternatives?

- (a) Diving (b) Driving
(c) Swimming (d) Sailing

Directions (Q. Nos. 70-72): In each question below are given two statements followed by two conclusions numbered I and II. You have to take the given two statements to be true even if they seem to be at variance from commonly known facts. Read the conclusions and then decide which of the following given conclusions logically follows from the two given statements, disregarding commonly known facts.

- 70. Statements** No women teacher play. Some women teachers are athletes.

Conclusions

- I. Male athletes can play. II. Some athletes can play.
(a) Neither I nor II follows (b) Only Conclusion I follows
(c) Either I or II follows (d) Only Conclusion II follows



71. **Statements** All mangoes are golden in colour. No golden coloured things are cheap.

Conclusions

I. All mangoes are cheap.

II. Golden-coloured mangoes are not cheap.

(a) Neither I nor II follows (b) Only Conclusion II follows

(c) Only Conclusion I follows (d) Either I or II follows

72. **Statements** All young scientists are open-minded. No open-minded. No open-minded men are superstitions.

Conclusions

I. No scientist is superstitious

II No young people are superstitious.

(a) Either I or II follows (b) Neither I nor II follows

(c) Only Conclusion II follows (d) Only Conclusion I follows

73. Six books are labelled A, B, C, D, E and F are placed side-by-side. Books B, C, E and F have green covers while others have yellow covers. Book A, B and D are new while the rest are old volumes. Book A, B and C are law reports while the rest are medical extracts. Which two books are old medical extracts and have green covers?

(a) B and C (b) E and F (c) C and F (d) C and E

74. Find out the wrong number in the following series. 30, -5, -45, -90, -145, -195, -255.

(a) -145 (b) -255 (c) -5 (d) -195

75. Find out the wrong number in the following series.

2, 5, 10, 17, 26, 38, 50, 65

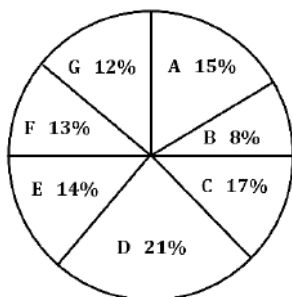
(a) 26 (b) 38 (c) 65

(d) 50

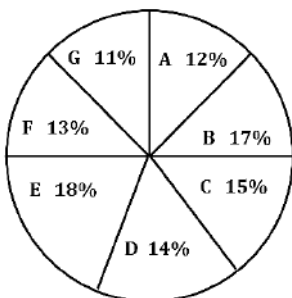
Directions (Q. Nos. 76-78) The following questions are based on the pie-charts given below:

Percentage-wise distribution of students studying in Arts and Commerce in seven different institutions, different institutions- A, B, C, D, E, F and G.

Total number of students studying Arts= 3800



Total number of students studying commerce = 4200



76. What is the total number of students studying Arts in Institutes A and G together?

(a) 1206 (b) 1226 (c) 1026 (d) 1126

77. How many students from Institute B study Arts and Commerce?

(a) 1108 (b) 1180 (c) 1018 (d) 1208

78. The ratio of the number of students studying Arts to that studying Commerce in institute E is

(a) 27: 14 (b) 19:28 (c) 19:27 (d) 19:16

79. **Statement** Many shops in the local market have extended their shops and occupied most part of the footpath in front of their shops.

Cause of Action

I. The civic authority should immediately activate a task force to clear all the footpaths encroached by the shop owners.

II. The civic authority should charge hefty penalty to the shop owners for occupying the footpath.

III. The civic authority should setup a monitoring system so that encroachments do not recur in future.

(a) None follows (b) II and III follow

(c) I and II follow (d) All I, II and III follow

80. **Statement** There is a significant increase in the number of patients affected by some disease in a city.

Cause of Action

I Municipal corporation of the city should take immediate action.

II This problem should be raised in the UNESCO.

III Hospitals in the city should be equipped properly for the treatment of the patients.

(a) Only III follows

(b) All follow

(c) I and II follow

(d) I and III follow

Directions (Q. Nos. 81-85) : Read the following information carefully and answer the questions.

Fives dramas A, B, C, D and E have to be staged in 6th where 1h needs to be given per drama.

1. A break of 1h has to be taken in 3rd or 4th hour.
2. Drama show cannot be started with A and cannot end C.
3. D has to follows B immediately with no break in between.
4. A cannot be done immediately after D.
5. A has to precede E immediately with no break in between.

81. Which hour is a break hour?

(a) 5th (b) 2nd (c) 3rd (d) 4th

82. Which is the drama to be staged first?

(a) A (b) D (c) B (d) None

83. Which is the drama staged immediately after the break?

(a) A (b) D (c) B (d) None

84. Which drama is staged immediately after D?

(a) None (b) E (c) B (d) C

85. Which drama is staged immediately after E?

(a) C (b) A (c) E (d) None

86. A person's present age two-fifth of the age of his mother. After 8 yr, he will be one-half of the age of the age of his mother. What is the present age of the mother?

(a) 40 yr (b) 60 yr (c) 30 yr (d) 50 yr



87. Running at the same constant rate, 6 identical machines can produce a total of 270 bottles per minute. At this rate, how many bottles could 10 such machines produce in 4 min?
(a)2700 (b)648 (c)10800
(d)1800
88. At what time, in minutes, between 3 o' clock and o' clock both the needles will coincide each other?
(a) $5\frac{1}{11}$ (b) $12\frac{4}{11}$ (c) $13\frac{4}{11}$ (d) $16\frac{4}{11}$
89. The greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5, respectively is
(a)127 (b)123 (c)235
(d)305
90. 1. A is the brother of B. 2. C is the father of A.
3. D is brother of E. 4. E is the daughter of B.
Then, The uncle of D is
(a)B (b)C (c)E (d)A

COMPUTER AWARENESS

91. The Boolean expression $AB+AB'+A'C+AC$ is unaffected by the value of the Boolean variable
(a)C (b)A (c)B (d)None
92. Suppose the largest n bit number requires, 'd' digits in decimal representation. Which of the following relations between 'n' and 'd' approximately correct?
(a) $d = 2^n$ (b) $n = 2^d$ (c) $d < n \log_{10} 2$ (d) $d > n \log_{10} 2$
93. FFFF will be the last memory location in a memory is size
(a)32k (b)16k (c)64k
(d)1k
94. Which of the following is equivalent to the Boolean expression $(X+Y) \cdot (X + \bar{Y}) \cdot (\bar{X} + Y)$
(a)XY (b) $\bar{Y}X$ (c) $\bar{X}Y$ (d) $\bar{X}\bar{Y}$
95. 'Floating point representation' is used to represent
(a)Integers (b)Real number
(c)Boolean values (d)Whole numbers
96. If a processor clock is rated as 2500 million cycles per second, then its clock period is
(a) 2.50×10^{-10} sec (b) 4.00×10^{-10} sec
(c) 1.00×10^{-10} sec (d)None of these
97. The minimum number of NAND gates required for implementing of Boolean expression, $AB + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}$ is
(a)1 (b)0 (c)2 (d)3
98. The maximum and minimum value represented in signed 16 bit 2's complement representation are
(a)-16384 and 16383 (b)0 and 32767
(c)-32768 and 32767 (d)0 and 65535
99. Write the simplified form of the Boolean expression $(A+C)(A+AD)+AC+C$
(a) $A+C'$ (b) $A+C$ (c) $A'+C$
(d) $A+D$
100. If a signal passing through a gate is inhibited by sending a low into one of the inputs, and the output is high, the gate is a(n)
(a)OR (b)NOR (c)NAND (d)AND

GENERAL ENGLISH

101. 'Bite and bullet' means
(a)to stop a conflict
(b)to become mad
(c)to accept something that is difficult or unpleasant
(d)to analyse your faults
102. What can you call a person who leads an unconventional style or living?
(a)Cynic (b)Bohemian
(c)Altruist (d)Agnostic
103. Fill in the blanks with the correct option.
Technical writing demandsuse of language.
(a)poetic (b)factual (c)figurative (d)dramatic
104. The county cleared this path and paved it with packed gravel, so they have a peaceful place to hike and bike.
Which of the following alternatives to the underlined portion would NOT be acceptable?
(a)Path, paving (b)Path and then paved
(c)Path before paving (d)Path paved
105. Select the correct form of verb/subject verb Agreement.
The principal, along with his assistants,..... the meeting
(a)are attending (b)attend
(c)is attending (d)attending
106. Select the correct articles.
He isM.A. with PhD and teaches inUniversity.
(a)the, the (b)an, the
(c)a, an (d)a, the
107. Fill in blank with the most appropriate option. Kedar.....this project for a month and now he is about to join a new project.
(a)guides (b)guided
(c)has been guiding (d)guiding

Comprehension type Questions:

Direction (Q.Nos. 108-110) Read the following passage carefully and answer the questions:

You might think you've experienced VR, and you might have been pretty impressed. Particularly, if you're a gamer, there are some great experiences to be had out there(or rather, in there)today. But over the next few year, in VR as in all fields of technology, we're going to see things that make what is cutting-edge today look like space invaders. And although the games will be amazing, the effects of this transformation will be far broader, touching on our work, education, and social lives.

Today's most popular VR applications involve taking total control of user's sense (sight and hearing, particularly) to create a totally immersive experience that places the user in a fully virtual environment that feels pretty realistic. Climb up something high and look down, and you're likely to get a sense of vertigo. If you see an object moving quickly towards your head, you'll feel an urge to duck out of the way.

Very soon, VR creators will extend this sensory hijacking to our other facilities-for example, touch and smell to deepen that sense of immersion. At the same time, the devices we use to visit these



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virtual worlds will become cheaper and lighter, removing the friction that can currently be a barrier.

I believe Extended Reality (XR)-a term that covers Virtual Reality (VR), Augmented Reality (AR), and Mixed reality (MR)-will be one of the most transformative tech trends of the next five years. It will be enabled and augmented by other tech trends, including super-fast networking, that will let us experience VR as a cloud service just like we currently consume music and movies. And Artificial Intelligence (AI) will provide us with more personalised virtual worlds to explore, even giving us realistic virtual characters to share our experiences with.

108.The passage states all the following about VR application expert.

- (a) Future AI will allow us to share our experiences with realistic virtual characters.
- (b) Vertigo is a major feature of all AI application.
- (c) VR application creates a virtual environment that feels pretty realistic.
- (d) VR applications take control of the user's senses.

109. 'Duck out of something' means

- (a) to avoid doing something
- (b) to fall down
- (c) to hit something hard
- (d) to meet with an accident

110. Select an antonym for the word 'argument' from the options given below.

- (a) Aggrandize
- (b) Curtail
- (c) Inflate
- (d) Reinforce

111. Read the following sentence to find if there is any error in any part.

If i were him/i would teach/him a lesson/no error

- (a) him a lesson
- (b) no error
- (c) if I were him
- (d) I would teach

112. Fill in the blanks with the correct preposition.

Jugdish is waiting for methe campus.

- (a) on
- (b) at
- (c) in
- (d) out

113. Select the word which means the same as the following.

To read something carefully

- (a) Presume
- (b) Perverse
- (c) Peruse
- (d) Persiflage

114. Select the one which best expresses the same sentence in indirect/direct speech:

He said, "I am glad to be here this evening"

- (a) He says he was glad to be here this evening
- (b) He said he was glad to be here this evening.
- (c) He said that he was glad to be there that evening.
- (d) He asked he is glad to be here this evening.

115. Choose the antonym.

BOLD

- (a) Nervous
- (b) Timid
- (c) Fearful
- (d) Coy

116. Which of the phrases given below should replace the phrase printed in bold to make the sentence grammatically correct.

He is addicted to smoke.

- (a) addicted with smoking
- (b) addict of smoke
- (c) used to smoke
- (d) addicted to smoking

117. Choose the correct alternative with the correct choice given below each statement.

You are to conform.....the rules of the Institute.

- (a) with
- (b) to
- (c) on
- (d) of

118. Choose the synonym.

LIBERAL

- (a) Generous
- (b) Affectionate
- (c) Sober
- (d) Reactionary

119. Identify the word that is similar in meaning to the underlined word.

Raghu make adulatory remarks about the waiter who served the food.

- (a) Complimentary
- (b) Ironic
- (c) Derogatory
- (d) Slanderous

120. Fill in the blank.

Neither Peter nor Iresponsible for this blunder.

- (a) am
- (b) are
- (c) is
- (d) were



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ANSWER KEY

1. (a)	21. (c)	41. (b)	61. (a)	81. (c)	101. (c)
2. (b)	22. (a)	42. (a)	62. (a)	82. (c)	102. (b)
3. (d)	23. (b)	43. (a)	63. (c)	83. (d)	103. (b)
4. (d)	24. (c)	44. (a)	64. (a)	84. (a)	104. (d)
5. (a)	25. (c)	45. (a)	65. (d)	85. (d)	105. (c)
6. (b)	26. (d)	46. (b)	66. (b)	86. (a)	106. (b)
7. (a)	27. (c)	47. (d)	67. (a)	87. (d)	107. (c)
8. (a)	28. (c)	48. (b)	68. (d)	88. (d)	108. (b)
9. (b)	29. (d)	49. (b)	69. (b)	89. (a)	109. (a)
10. (b)	30. (c)	50. (d)	70. (a)	90. (d)	110. (b)
11. (c)	31. (b)	51. (b)	71. (b)	91. (c)	111. (b)
12. (d)	32. (b)	52. (a)	72. (b)	92. (d)	112. (a)
13. (c)	33. (c)	53. (d)	73. (b)	93. (c)	113. (c)
14. (d)	34. (a)	54. (d)	74. (a)	94. (a)	114. (c)
15. (a)	35. (c)	55. (d)	75. (b)	95. (b)	115. (b)
16. (a)	36. (c)	56. (a)	76. (c)	96. (b)	116. (d)
17. (d)	37. (b)	57. (b)	77. (c)	97. (b)	117. (b)
18. (b)	38. (c)	58. (a)	78. (c)	98. (c)	118. (a)
19. (a)	39. (b)	59. (a)	79. (d)	99. (b)	119. (a)
20. (c)	40. (c)	60. (a)	80. (d)	100. (c)	120. (a)

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SOLUTION

MATHS

1. (a) Note that area of the parallelogram bounded by the lines $y = m_1x + c_1$, $y = m_1x + c_2$ and $y = m_2x + d_1$, $y = m_2x + d_2$ is given by $\left| \frac{(c_1 - c_2)(d_1 - d_2)}{m_1 - m_2} \right|$

Area of the parallelogram

$$y = 4x + 0 \quad y = 4x + 1$$

$$y = -x + 0 \quad y = -x + 1$$

$$c_1 = 0, c_2 = 1 \quad d_1 = 0, d_2 = 1$$

$$m_1 = 4, \quad m_2 = -1$$

$$\left| \frac{(c_1 - c_2)(d_1 - d_2)}{m_1 - m_2} \right| = \left| \frac{(0 - 1)(0 - 1)}{4 - (-1)} \right| = \frac{1}{5}$$

2. (b) The function $f(x) = \begin{cases} (1 + 2x)^{1/x}, & x \neq 0 \\ e^2, & x = 0 \end{cases}$

Check continuity

at $x = 0$

$$\text{LHL} = \text{RHL} = f(0)$$

$$\Rightarrow \lim_{x \rightarrow 0} (1 + 2x)^{1/x} = e^2$$

$$\Rightarrow e^{\lim_{x \rightarrow 0} (1 + 2x - 1) \times \frac{1}{x}} = e^2$$

$$\left[\begin{array}{l} \text{Use formula } \lim_{x \rightarrow a} (1 + f(x))^{g(x)} \\ \lim_{x \rightarrow 0} (1 + f(x) - 1)g(x) \\ \text{then, } e^{x \rightarrow 0} \end{array} \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2x}{x} = e^2$$

$$\Rightarrow e^2 = e^2$$

$f(x)$ is continuous at $x = 0$.

For differentiability:

$$f(x) = (1 + 2x)^{1/x}$$

$$\log f(x) = \frac{1}{x} \log(1 + 2x)$$

Differentiate both side

$$\frac{1}{f(x)} f'(x) = \frac{2}{x(1+2x)} + \log(1 + 2x) \left(\frac{-1}{x^2} \right)$$

$$f'(x) = (1 + 2x)^{1/x} \left[\frac{2}{x(1+2x)} - \frac{\log(1+2x)}{x^2} \right]$$

$f'(x)$ is undefined at $x = 0$.

$\Rightarrow f(x)$ is not differentiable at $x = 0$

3. $x^m \cdot y^n > (x + y)^{m+n}$

$$m \log x + n \log y > (m + n) \log (x + y)$$

differentiate w.r.t. x

$$\frac{m}{x} + \frac{n}{y} \frac{dy}{dx} = \frac{m+n}{x+y} \left(1 + \frac{dy}{dx} \right)$$

$$\frac{dy}{dx} \left(\frac{n}{y} - \frac{m+n}{x+y} \right) = \frac{m+n}{x+y} - \frac{m}{x}$$

$$\frac{dy}{dx} \left(\frac{nx + ny - my - ny}{y(x+y)} \right) = \frac{mx + nx - mx - my}{x(x+y)}$$

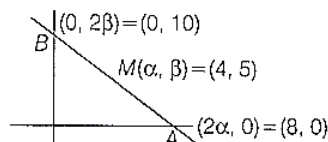
$$\frac{dy}{dx} > \left(\frac{nx - my}{nx - my} \right) \frac{y}{x}$$

$$\frac{dy}{dx} = \frac{y}{x} \quad (d)$$

4. (d)

If (α, β) be the mid point of intercepted part AB on axes, then $A(2\alpha, 0)$, $B(0, 2\beta)$ equation of AB will be $\frac{x}{2\alpha} + \frac{y}{2\beta} = 1$.

Applying same concept



$$\frac{x}{8} + \frac{y}{10} = 1 \Rightarrow 5x + 4y = 40$$

5. (a) $\begin{vmatrix} \lambda & 1 & -2 \\ 1 & \lambda & -2 \\ 1 & 1 & 1 \end{vmatrix} = 7$
- $$\lambda(x+2) - 1(1+2) - 2(1-\lambda) = 7$$
- $$\lambda^2 + 4x - 12 = 0$$
- $$(\lambda - 2)(\lambda + 6) = 0$$
- $$\lambda = 2, -6$$

6. (b) $3^3 \times 3 - \log_3 5 = 3^3 \cdot 3^{\log_3 5^{-1}} = 3^3 \times 5^{-1} = \frac{27}{5}$

7. (a) We have

$$d = a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \dots = a_n - a_{n-1}$$

Thus, $\sin d [\operatorname{cosec} a_1 \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \operatorname{cosec} a_3$

$$+ \dots + \operatorname{cosec} a_{n-1} \operatorname{cosec} a_n]$$

$$= \frac{\sin(a_2 - a_1)}{\sin a_1 \sin a_2} + \frac{\sin(a_3 - a_2)}{\sin a_2 \sin a_3} + \dots + \frac{\sin(a_n - a_{n-1})}{\sin a_{n-1} \sin a_n}$$

$$= (\cot a_1 - \cot a_2) + (\cot a_2 - \cot a_3) + \dots +$$

$$= \cot a_1 - \cot a_n (\cot a_{n-1} - \cot a_n)$$

Hence, option (a) is the correct answer.

8. $\operatorname{cosec}^{-1} \frac{5}{3} \Rightarrow \tan^{-1} \frac{3}{4}$

$$\cot \left[\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{2}{3} \right]$$

$$\cot \left[\tan^{-1} \left[\frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \cdot \frac{2}{3}} \right] \right]$$

$$\cot \left[\cot^{-1} \left[\frac{6}{17} \right] \right]$$

$$= \frac{6}{17}$$

9. (b) Let $a_1, a_2, a_3, \dots, a_n$ are n positive numbers

AM of m^{th} power of numbers $\geq (\text{AM})^m$

$$\frac{a_1^m + a_2^m + a_3^m + \dots + a_n^m}{n} \geq \left(\frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \right)^m$$

$$m = 2$$

$$\frac{a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2}{n} \geq \left(\frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \right)^2$$

$$\Rightarrow n \sum_{i=1}^n a_i^2 \geq \left(\sum_{i=1}^n a_i \right)^2$$

10. (b) $\cos^{-1} \frac{x}{2} + \cos^{-1} \frac{y}{3} = \phi$

$$\cos \left(\cos^{-1} \frac{x}{2} + \cos^{-1} \frac{y}{3} \right) = \cos \phi$$

$$\cos \left[\cos^{-1} \left(\frac{x}{2} \right) \right] \cos \left[\cos^{-1} \left(\frac{y}{3} \right) \right]$$

$$- \sin \cos^{-1} \frac{x}{2} \sin \cos^{-1} \frac{y}{3} = \cos \phi$$



$$\frac{x}{2} \cdot \frac{y}{3} - \sin\left(\sin^{-1}\sqrt{1-\frac{x^2}{4}}\right) \sin\left(\sin^{-1}\sqrt{1-\frac{y^2}{9}}\right) = \cos \phi$$

$$\frac{xy}{6} - \sqrt{1-\frac{x^2}{4}} \sqrt{1-\frac{y^2}{9}} = \cos \phi$$

$$\frac{xy}{6} - \cos \phi = \sqrt{1-\frac{x^2}{4}} \sqrt{1-\frac{y^2}{9}}$$

$$\frac{x^2 y^2}{36} + \cos^2 \phi - \frac{xy \cos \phi}{3} = \left(1 - \frac{x^2}{4} - \frac{y^2}{9} + \frac{x^2 y^2}{36}\right)$$

$$9x^2 + 4y^2 - 12xy \cos \phi = 36 \sin^2 \phi$$

11. (c) $x^2 + px + q = 0$ has roots are $\tan 30^\circ$ and $\tan 15^\circ$

$$\tan 30^\circ + \tan 15^\circ = -p$$

$$\tan 30^\circ \tan 15^\circ = q$$

$$\tan 45^\circ = \frac{\tan 30^\circ + \tan 15^\circ}{1 - \tan 30^\circ \tan 15^\circ} = \frac{-p}{1-q} = 1$$

$$-p = 1 - q \Rightarrow -1 = p - q$$

$$2 - 1 = 2 + p - q = 1$$

12. (d) $\operatorname{cosec} \theta - \cot \theta = 2$

$$\operatorname{cosec} \theta + \cot \theta = \frac{1}{2}$$

$$\text{Adding Eqs. (i) and (ii) we get } 2 \operatorname{cosec} \theta = \frac{5}{2}$$

$$\text{Hence, } \operatorname{cosec} \theta = \frac{5}{4}$$

13. $\tan^{-1} \sqrt{x^2 + x} = \frac{\pi}{2} - \sin^{-1} \sqrt{x^2 + x + 1}$

$$\Rightarrow \tan^{-1} \sqrt{x^2 + x} = \cos^{-1} \sqrt{x^2 + x + 1}$$

$$\Rightarrow \cos^{-1} \frac{1}{\sqrt{x^2 + x + 1}} = \cos^{-1} \sqrt{x^2 + x + 1} \text{ as}$$

$$\tan^{-1} x \Rightarrow \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$$

$$\frac{1}{\sqrt{x^2 + x + 1}} \cdot \sqrt{x^2 + x + 1}$$

$$x^2 + x + 1 = 1$$

$$x = 0, -1$$

14. (d) $\int_a^b |x-a| dx + \int_a^b |x-b| dx$ as $a < x < b$

$$= \int_a^b (x-a) dx - \int_a^b (x-b) dx$$

$$= b \int_a^b dx - a \int_a^b dx$$

$$= (b-a)^2$$

15. Let $\vec{b} = xi + 2j + 2\vec{k}$

$$a \cdot b = 14$$

$$2x + 2y + z = 14 \quad \dots\dots\dots(1)$$

$$a \times b = 3i - j - 8\vec{k}$$

$$a \times b = \begin{vmatrix} i & j & \vec{k} \\ 2 & 2 & 1 \\ x & y & z \end{vmatrix} (2z-y)i - (2z-x)j + (2z-2x)\vec{k}$$

$$3i - j - 8\vec{k} = (2z-y)i - (2z-x)j + (2z-2x)\vec{k}$$

$$2z-y = 3, 2z-x = -1, 2z-2x = -8$$

$$\text{From equation } \dots\dots\dots(1)$$

$$X = 5, y = 1, z = 2$$

$$[5i + j + 2\vec{k}] (a)$$

16. (a) Distance between parallel lines

$$ax + by + c_1 = 0, ax + by + c_2 = 0 \Rightarrow d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$$

Note For applying this formula coefficient of x and y must be same in both lines.

$$-2x + y = 2 \text{ and } 2x - y = 2$$

$2x - y + 2 = 0$ and $2x - y - 2 = 0$ as we know coefficient of x and y must be same in both lines.

$$a = \frac{|2 - (-2)|}{\sqrt{2^2 + (-1)^2}} = \frac{4}{\sqrt{5}}$$

$$4x - 3y - 5 = 0 \text{ and } 8x - 6y + 1 = 0$$

$$4x - 3y - 5 = 0 \text{ and } 4x - 3y + \frac{1}{2} = 0$$

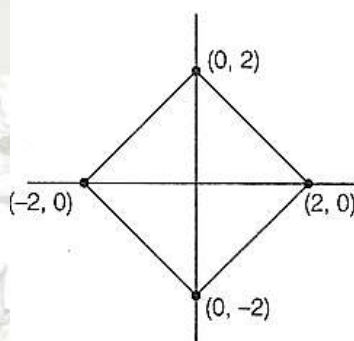
$$b = \frac{|(-5) - \frac{1}{2}|}{\sqrt{4^2 + (-3)^2}} = \frac{11}{10} \Rightarrow a = \frac{4}{\sqrt{5}}, b = \frac{11}{10}$$

$$10b = 11 \Rightarrow 40b = 44$$

$$\Rightarrow 40b = 4 \times 11 \Rightarrow 40b = \sqrt{5}a \times 11$$

17. (d) $|x| + |y| = 2 \rightarrow \text{Area}$

$$\text{Area} = l \times b = 2\sqrt{2} \times 2\sqrt{2} = 8 \text{ sq. units}$$



Short Trick:

The area bounded by

$$|x| + |y| = a, a > 0$$

$$\text{Area} = (\sqrt{2}a)^2 \text{ sq. unit}$$

In this question $a = 2$

$$\text{Hence, area} = (\sqrt{2} \times 2)^2 = 2 \times 4$$

$$= 8 \text{ sq. units}$$

18. (b) H.P. A.P.

$$T_p = q T_p = 1/q = A + (p-1)d$$

$$T_q = p T_q = 1/p = A + (q-1)d$$

$$\frac{1}{q} - \frac{1}{p} = (p-q)d \Rightarrow d = \frac{1}{pq}$$

Put this value in Eq. (ii)

$$\frac{1}{p} = A + (q-1) \frac{1}{pq}$$

$$= A + \frac{1}{p} - \frac{1}{pq}$$

$$\Rightarrow A = \frac{1}{pq}$$

$$T_{pq} = A + (pq-1)d = \frac{1}{pq} + (pq-1) \times \frac{1}{pq} = 1$$



19. (a) The temperature function

$$T(x, y) = 4x^2 - 4xy + y^2$$

$$= (2x - y)^2$$

The ant traverses the circle of radius and center at origin

$$\text{Let } 2x - y = K$$

Then K will be the maximum when line $2x - y = K$ touches the circle.

$$K^2 = 25(1 + 2^2) = 125$$

20. (c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$

$$\frac{dy}{dx} = -\frac{x b^2}{y a^2}$$

$$\text{and } x^2 - y^2 = c^2$$

$$2x - 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{x}{y}$$

For orthogonal intersection $m_1 \times m_2 = -1$

$$\frac{dy}{dx} \times \frac{dy}{dx} \Rightarrow -\frac{b^2 x}{a^2 y} = -1$$

$$b^2 x^2 = a^2 y^2 \Rightarrow \frac{x^2}{a^2} = \frac{y^2}{b^2}$$

Put this value in ellipse, we get

$$\frac{2x^2}{a^2} = 1 \Rightarrow x^2 = \frac{a^2}{2} \Rightarrow x = \pm \frac{a}{\sqrt{2}}$$

Hence,

$$y^2 = \frac{b^2}{2} \Rightarrow y = \pm \frac{b}{\sqrt{2}}$$

Put these values in hyperbola, we get

$$x^2 - y^2 = c^2$$

$$\frac{a^2}{2} - \frac{b^2}{2} = c^2$$

Hence,

$$a^2 - b^2 = 2c^2$$

OR

If ellipse and hyperbola are orthogonal it means they are confocal it means they have same coordinates of focus.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b, e = \frac{\sqrt{a^2 - b^2}}{a}$$

Its foci are $(\pm ae, 0)$

And foci of $x^2 - y^2 = c^2$ are $(\pm \sqrt{2}c, 0)$

$$\frac{a\sqrt{a^2 - b^2}}{a} = \sqrt{2}c \Rightarrow a^2 - b^2 = 2c^2$$

21. (c) $(\hat{a} \times \hat{b}) \cdot \hat{c} = \hat{a} \cdot (\hat{b} \times \hat{c})$

$$(\hat{c} \cdot \hat{a}) \hat{b} - (\hat{c} \cdot \hat{b}) \hat{a} = (\hat{a} \cdot \hat{c}) \hat{b} - (\hat{a} \cdot \hat{b}) \hat{c}$$

$$(\hat{c} \cdot \hat{b}) \hat{a} = (\hat{a} \cdot \hat{b}) \hat{c} \quad \left\{ \begin{array}{l} -\hat{c} \times (\hat{a} \times \hat{b}) \\ -[(\hat{c} \cdot \hat{b}) \hat{a} - (\hat{c} \cdot \hat{a}) \hat{b}] \\ (\hat{c} \cdot \hat{a}) \hat{b} - (\hat{c} \cdot \hat{b}) \hat{a} \end{array} \right\}$$

$$\lambda \hat{a} = \mu \hat{c}$$

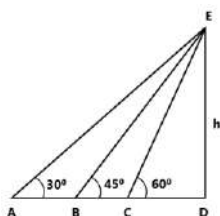
$$a \parallel c$$

$\hat{a}$ and $\hat{c}$ are collinear.

22. (a) Let $DE = h$

$$\tan 30^\circ = \frac{h}{AP}, \text{AO} = \frac{h}{\tan 30}$$

$$AD = h\sqrt{3}$$



$$BD \cdot \frac{h}{\tan 45^\circ} \Rightarrow h$$

$$CD = \frac{h}{\tan 60} \Rightarrow \frac{h}{\sqrt{3}}$$

$$= \frac{\sqrt{3}}{1}$$

$$AB : BC = \sqrt{3} : 1$$

23. (b) All vectors line in a plane.

$$\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$$

Expand along R_2 ,

$$\Rightarrow -1(ab - c^2) + 0(ab - c^2) - 1(ac - ac) = 0$$

$$\Rightarrow -ab + c^2 = 0 \Rightarrow c^2 = ab$$

$a, c, b \rightarrow G.P.$

$\therefore c$ is geometric mean of a and b .

24. (c) $n(A) = m; n(B) = n$

$$|P(A)| - |P(B)| = 112$$

$$\Rightarrow 2^m - 2^n = 112$$

$$\Rightarrow m = 7; n = 4$$

$$2m - n = 1$$

$$14 - 4 \neq 1$$

25. (c)

$$\text{Mean} = \frac{x_1 + x_2 + x_3 + \dots + x_{25}}{25} = 38$$

$$x_1 + x_2 + x_3 + \dots + x_{25} = 38 \times 25$$

23 is misread as 25 and 38 misread 36.

There is difference of $25 - 23 = 2$

and difference of $36 - 38 = -2$ in the sum.

Therefore there is no any difference in the net sum during calculation of mean.

Hence, the mean remain constant.

26. (d) $x^2 + y^2 - 2y = 0$ centre $(0,1)$ radius $(r) = 1$

$$x^2 + y^2 - 2y - 3 = 0 \text{ centre } (0,1) \text{ radius } (R) = 2$$

$$\text{Sample space } \pi R^2 = \pi \times 2^2 = 4\pi$$

Then, the probability that the point has been taken from smaller circle is

$$\text{Favourable cases} = \pi r^2 = \pi \times 1^2 = \pi$$

$$\text{Probability} = \frac{\pi}{4\pi} = \frac{1}{4}$$

27. (c)

$$\int \frac{(x^2-1)dx}{x^3 \sqrt{2x^4-2x^2+1}}$$

$$\Rightarrow \text{Let } I = \frac{x^2-1}{\sqrt{2x^4-2x^2+1}}$$

$$I = \frac{x^2-1}{x^5 \sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}} dx$$

$$I = \frac{\frac{1}{x^3} - \frac{1}{x^5}}{\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}} dx$$



$$\text{Let } 2 - \frac{2}{x^2} + \frac{1}{x^4} = +$$

$$\left(0 + \frac{4}{x^3} - \frac{4}{x^5}\right) dx = \frac{dt}{4}$$

$$\int \frac{dt}{4\sqrt{t}} = \frac{1}{4} \int t^{-\frac{1}{2}} dt = \frac{1}{4} \frac{t^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + c$$

$$\frac{1}{4} \times \frac{t^{\frac{1}{2}}}{\frac{1}{2}} \times 2 + c = \frac{1}{2} t^{\frac{1}{2}} + c$$

$$\sqrt{\frac{2 - \frac{2}{x^2} + \frac{1}{x^4}}{2}} + c$$

$$\frac{1}{2} \sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + c$$

- 28. (c)** $P(A \cap B) = P(A)P(B)$, hence A and B are independent events.

Option (a) $P\left(\frac{B}{A}\right) = P(B)$ and option

(d) $P\left(\frac{A}{B}\right) = P(A)$

(b) $P(A \cap B^c) = P(A \cap B) + P(B^c) - P(A \cap B)$

Hence, (a) and (d) options are wrong.

Option (c) $P(\overline{A \cup B}) = P(\overline{A} \cap \overline{B}) = P(\overline{A}) \cdot P(\overline{B})$

Hence, option (c) is correct.

- 29. (d)** (a) $\lim_{x \rightarrow \infty} \frac{x}{e^x} = \frac{\infty}{\infty}$ (from ∞)

Using L' Hospital rule

$$\lim_{x \rightarrow \infty} \frac{1}{e^x} = \frac{1}{e^\infty} = \frac{1}{\infty} = 0$$

Hence, it is a right answer.

$$(b) \lim_{x \rightarrow 0^+} \frac{1/x}{e^{1/x}}$$

Let $\frac{1}{x} = t \rightarrow \infty = \lim_{t \rightarrow \infty} \frac{t}{e^t}$ same question as in option (a).

Hence it is also right answer.

$$(c) \lim_{x \rightarrow 0^+} \frac{\sin x}{1+2x} = \lim_{x \rightarrow 0^+} \frac{(\sin x)/x}{(1/x)+2} = \frac{1}{\infty+2} = 0$$

Hence, it is also right answer.

$$(d) \lim_{x \rightarrow 0^+} \frac{\cos x}{1+2x} = \frac{1}{1+0} = 1.$$

Hence, this is the wrong answer.

- 30. (c)** A four digit number is formed using the digits 1, 2, 3, 4, 5 without repetition
Sample space = ${}^5P_4 = 5! = 120$
The probability that number is divisible by 3 .
Any number divisible by 3 if sum of its digit is divisible by 3 .

Hence, if we choose 4 digits 1, 2, 4, 5.

Sum of its digits is 12 . Favourable cases = 4 !

$$\text{Probability} = \frac{4!}{5!} = \frac{1}{5}$$

- 31. (b)**

$$\text{Let } y = \cos^{-1}(-x)$$

$$\cos y = -x$$

$$-\cos y = x$$

As we know

$$\cos(\pi - y) = -x$$

$$\pi - y = \cos^{-1}x$$

$$\pi - \cos^{-1}x = y$$

$$\text{Hence, } y = \cos^{-1}(-x) = \pi - \cos^{-1}x$$

- 32. (b)** If α and β are the roots of

$$x^2 - x - 1 = 0$$

$$\alpha + \beta = 1, \alpha\beta = -1$$

$$A_n = \alpha^n + \beta^n$$

$$\text{AM of } A_{n-1} \text{ and } A_n = \frac{A_{n-1} + A_n}{2}$$

$$= \frac{\alpha^{n-1} + \beta^{n-1} + \alpha^n + \beta^n}{2}$$

$$= \frac{\alpha^{n-1}(1+\alpha) + \beta^{n-1}(1+\beta)}{2}$$

As α, β are roots of $x^2 - x - 1 = 0$

$$\text{Hence, } \alpha^2 - \alpha - 1 = 0 \Rightarrow 1 + \alpha = \alpha^2 \text{ and } 1 + \beta = \beta^2$$

$$\frac{\alpha^{n-1}(\alpha^2) + \beta^{n-1}\beta^2}{2} = \frac{\alpha^{n+1} + \beta^{n+1}}{2} = \frac{A_{n+1}}{2}$$

$$\mathbf{33. (c)} D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2+x & 1 \\ 1 & 1 & 2+y \end{vmatrix}$$

$$= [(x+2)(y+2) - 1] - [y+2-1] + 1[1-x-2]$$

$$= (xy + 2x + 2y + 4 - 1) - y - 1 + 1 - x - 2$$

$$= xy + x + y + 1$$

$$= (1+x)(1+y)$$

$$\mathbf{34. (a)} \tan\left(\frac{B-C}{2}\right) = \frac{1}{3} \tan\left(\frac{B+C}{2}\right)$$

$$\text{Using Napier's analogy, } \tan\left(\frac{B-C}{2}\right) = \left(\frac{b-c}{b+c}\right) \cot \frac{A}{2}$$

$$\text{As we know, } \tan\left(\frac{B+C}{2}\right) = \cot\left(\frac{A}{2}\right)$$

$$= \frac{1}{3} \tan\left(\frac{B+C}{2}\right) = \frac{1}{3} \cot \frac{A}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$$

$$\text{Hence, } \frac{b-c}{b+c} = \frac{1}{3} \Rightarrow b = 2c \Rightarrow b:c = 2:1$$

- 35. (c)**

$$\text{Volume} = [abc] = \begin{vmatrix} 2 & 3 & 4 \\ 1 & a & 2 \\ 1 & 2 & a \end{vmatrix} = 15$$

$$2(a^2-5) + 3(2-a) + 4(2-a) = 15$$

$$2a^2 - 8 + 6 - 3a + 8 - 4a = 15$$

$$2a^2 - 7a - 9 = 0$$

$$2a^2 - 9a + 2a - 9 = 0$$

$$a(2a-9) + (2a-9) = 0$$

$$(2a-9)(a+1) = 0$$

$$(2a-9) = 0 \quad \text{or} \quad (a+1) = 0$$

$$a = 9/2, \quad a = -1$$



36. (c)

$$\mu_r = \frac{\Sigma(x-\bar{x})^r}{n}, \mu_r = r\text{th}$$

Moment about mean and $\Sigma(x - \bar{x}) = \text{Mean deviation}$

$$\mu_1 = \frac{\Sigma(x-\bar{x})}{n}; \mu_2 = \frac{\Sigma(x-\bar{x})^2}{n}; \mu_3 = \frac{\Sigma(x-\bar{x})^3}{n}$$

$$\mu_1 \text{ about } 2 \text{ is } 1, \mu_2 \text{ is } 16; \mu_3 = 40$$

$$\mu_1 = \frac{\Sigma(x_i - 2)}{n} = 1 \quad \mu_2 = \frac{\Sigma(x_i - 2)^2}{n} = 16$$

$$\Sigma_{i=1}^n (x_i - 2) = n \quad \Sigma(x_i^2 + 4 - 4x_i) = 16n$$

$$\Sigma_{i=1}^n x_i - 2 \Sigma_{i=1}^n 1 = n \quad \Sigma x_i^2 - 4 \Sigma x_i + 4n = 16n$$

$$\Sigma x_i = 3n \Rightarrow \text{mean } \frac{\Sigma x_i}{n} = 3 \quad \Sigma x_i^2 = 24n$$

$$\text{Variance} = (\text{SD})^2 \cdot V(x) = \frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2$$

$$= \frac{24n}{n} - (3)^2 = 24 - 9 = 15$$

37. (b) $y = \frac{10^x - 10^{-x}}{10^x + 10^{-x}} = \frac{10^{2x} - 1}{10^{2x} + 1}$

$$y \times 10^{2x} + y = 10^{2x} - 1$$

$$y + 1 = 10^{2x}(1 - y)$$

$$10^{2x} = \frac{1+y}{1-y} \Rightarrow \log_{10} 10^{2x} = \log_{10} \left(\frac{1+y}{1-y}\right)$$

$$\Rightarrow 2x = \log_{10} \left(\frac{1+y}{1-y}\right) \Rightarrow x = \frac{1}{2} \log_{10} \left(\frac{1+y}{1-y}\right)$$

$$x \leftrightarrow y$$

$$y = \frac{1}{2} \log_{10} \left(\frac{1+x}{1-x}\right) = f^{-1}(x)$$

38. (c) For focus of parabola $4(y^2 - 5y) = -12x - 67$

$$4\left(y^2 - 2\left(\frac{5}{2}\right)y + \frac{25}{4}\right) = -12x - 67 + 25 = -12x - 42$$

$$4\left(y - \frac{5}{2}\right)^2 = -12\left(x + \frac{7}{2}\right)$$

$$\Rightarrow \left(y - \frac{5}{2}\right)^2 = -3\left(x + \frac{7}{2}\right)$$

$$\left(y - \frac{5}{2}\right)^2 = -4\left(\frac{3}{4}\right)\left(x + \frac{7}{2}\right)$$

$$\text{Hence, vertex will be } \left(-\frac{7}{2}, \frac{5}{2}\right) \text{ and } a = \frac{1}{3}$$

Focus will be

$$\left(-\frac{7}{2} - \frac{3}{4}, \frac{5}{2}\right) \Rightarrow \left(-\frac{17}{4}, \frac{5}{2}\right)$$

39. (b) $t_1 = \frac{\sqrt{5}}{2} = \frac{\sqrt{5}}{9}$

$$t_2 = \frac{\sqrt{5}}{4} = \frac{\sqrt{5}}{\sqrt{16}}$$

$$t_3 = \frac{\sqrt{5}}{5} = \frac{\sqrt{5}}{\sqrt{25}} = \frac{1}{\sqrt{5}}$$

$$t_n = \frac{\sqrt{5}}{r+2}$$

$$\text{So, } r+2 = 13$$

$$r = 13 - 2$$

$$r = 11$$

40. (c) $a = x - x^2$, a is acceleration

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = x - x^2$$

Here, x is the distance and v is velocity.

$$\text{At } t = 0, u = 0$$

Where, u is initial velocity,

$$\frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt}$$

$$\Rightarrow x - x^2 = v \frac{dv}{dx}$$

$$\Rightarrow v \frac{dv}{dx} = x - x^2$$

On integrating Eq. (i)

$$\int v dv = \int (x - x^2) dx \Rightarrow \frac{v^2}{2} = \frac{x^2}{2} - \frac{x^3}{3}$$

For $v = 0$,

$$3x^2 - 2x^3 = 0 \Rightarrow x = 0, 3/2$$

Hence, answer is $x = 3/2$ because particle start from rest.

41. (b) $f(x) = \log(x + \sqrt{x^2 + 1})$

$$f(-x) = \log(-x + \sqrt{(-x)^2 + 1})$$

$$= f(x) + \epsilon(-x)$$

$$= \log[(x + \sqrt{1 + x^2})(-x + \sqrt{1 + x^2})]$$

$$= \log(-x^2 + (1 + x^2))$$

$$= \log 1 \Rightarrow 0$$

$$f(-x) = -f(x) \text{ is an odd function.}$$

42. (a) $\cos^{-1} x$

$$\text{Here, } -1 \leq x \leq 1$$

$$\text{Now, } [x] \neq 0$$

$$\text{Hence, } x \text{ can not lie between } 0 \leq x < 1.$$

$$\text{Domain of function is } (-1 \leq x < 0) \cup x = 1$$

$$x \in [-1, 0) \cup \{1\}$$

43. (a)

$$n(U) = 200$$

$$n(T) = \frac{60}{100} \times 200 = 120$$

$$n(C) = \frac{72}{100} \times 200 = 144$$

$$n(T \cap C) = x$$

$$m \leq x \leq n \quad n = \max$$

$$m = \min$$

$$n(T \cap C) \text{ is max when } T \subset C$$

In that case

$$n(T \cap C) = n(T) = n = 120$$

$$\text{Hence, } n = 120$$

$$n(T \cap C) \text{ is minimum when } n(T \cup C) = n(U)$$

$$n(T \cup C) = n(T) + n(C) - n(T \cap C)$$

$$200 = 120 + 144 - n(T \cap C)$$

$$n(T \cap C) = 64 = m$$

$$n - m = 120 - 64 = 56$$



44. (a)

$$\begin{aligned} 4\cos^2 x + 6\sin^2 x &= 5 \\ \Rightarrow 2\sin^2 x &= 1 \\ \Rightarrow \sin^2 x &= \frac{1}{2} = \sin^2 \frac{\pi}{4} \\ \Rightarrow x &= n\pi \pm \frac{\pi}{4} \end{aligned}$$

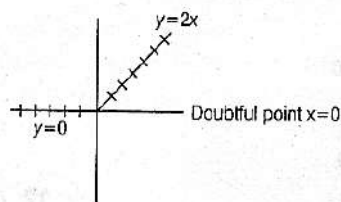
45. (a) $f(x) = x + |x|$ is continuous for

$$f(x) = \begin{cases} x + x = 2x, & x \geq 0 \\ x - x = 0, & x < 0 \end{cases}$$

$$\text{LHL} = \text{RHL} = f(0)$$

$$\lim_{x \rightarrow 0^-} = \lim_{x \rightarrow 0^+} 2x = f(0)$$

Hence, continuous at $x = 0$



Continuous at $x \in (-\infty, \infty)$

46. (b)

$$23, 29, 32, 49, 118$$

$$n = 5$$

$$P_i = \frac{i(n+1)}{100} = i \text{ th percentile}$$

$$P_{10} = \frac{10(n+1)}{100} = \frac{n+1}{10} = \frac{6}{10} = \frac{3}{5} \approx 1$$

$$P_{50} = \frac{50(n+1)}{100} = \text{median} = \frac{n+1}{2} = \frac{5+1}{2} = 3$$

$$\Rightarrow 23, 32$$

47. (d)

$$\lim_{n \rightarrow \infty} \frac{1^a + 2^a + \dots + n^a}{(n+1)^{a-1} \{ (na+1) + (na+2) + \dots + (na+n) \}} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{r^a}{(n+1)^{a-1} \left[n^2 a + \frac{n(n+1)}{2} \right]} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{2 \sum_{r=1}^n \left(\frac{r}{n} \right)^a}{\left(1 + \frac{1}{n} \right)^{a-1} \cdot (2na + n + 1)} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} \left(2 \sum_{r=1}^n \left(\frac{r}{n} \right)^a \right) \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n} \right)^{a-1} \cdot (2a + 1 + \frac{1}{n})} = \frac{1}{60}$$

$$= 2 \int_0^1 (x^a) dx \cdot \frac{1}{1 \cdot (2a+1)} = \frac{1}{60} = \frac{2 \cdot [x^{a+1}]_0^1}{(2a+1) \cdot (a+1)} = \frac{1}{60}$$

$$= 2 \int_0^1 (x^a) dx \cdot \frac{1}{1 \cdot (2a+1)} = \frac{1}{60} = \frac{2 \cdot [x^{a+1}]_0^1}{(2a+1) \cdot (a+1)} = \frac{1}{60}$$

$$\therefore \frac{2}{(2a+1)(a+1)} = \frac{1}{60}$$

$$\Rightarrow (2a+1)(a+1) = 120$$

$$\Rightarrow 2a^2 + 3a + 1 - 120 = 0$$

$$\Rightarrow 2a^2 + 3a - 119 = 0$$

$$\Rightarrow (2a+17)(a-7) = 0$$

$$\Rightarrow a = 7, \frac{-17}{2}$$

48. (b) $e = \frac{1}{3}$ directrix $x = 9 = \frac{a}{e}$

$$\text{Hence, } a = 3$$

$$b^2 = a^2(1 - e^2) = 9 \left(1 - \frac{1}{9} \right) = 8$$

$$\text{Equation of ellipse } \frac{x^2}{9} + \frac{y^2}{8} = 1$$

$$\Rightarrow 8x^2 + 9y^2 = 72$$

49. (b) $\frac{x^2}{25} + \frac{y^2}{b^2} = 1$ (Ellipse)

$$; \frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25} \text{ (Hyperbola)}$$

$$a = 5, \quad A = \frac{12}{5}, B = \frac{9}{5}$$

$$e^2 = 1 - \frac{b^2}{a^2} \quad e^2 = 1 + \frac{B^2}{A^2}$$

$$e^2 = 1 + \frac{81}{144} \Rightarrow \frac{15}{12} = e$$

Focus of ellipse : $(ae, 0)$ Focus of hyperbola : $(Ae, 0)$

$$(5e, 0) \left(\frac{12}{5} \times \frac{15}{12}, 0 \right) = (3, 0)$$

Given:

$$(5e, 0) = (3, 0)$$

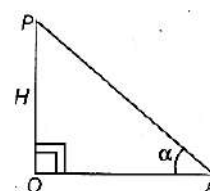
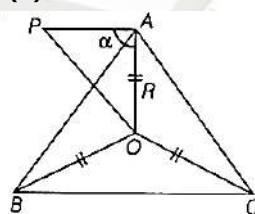
$$5e = 3 \Rightarrow e = \frac{3}{5}$$

$$\frac{9}{25} = 1 - \frac{b^2}{25}$$

$$\Rightarrow \frac{b^2}{25} = 1 - \frac{9}{25}$$

$$\text{Hence, } b^2 = 16$$

50. (d)



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R \quad \tan \alpha = \frac{H}{R}$$

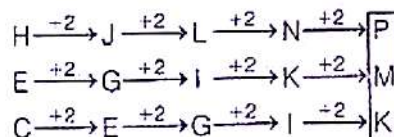
$$\tan \alpha = \frac{2H \sin A}{a}$$

$$\Rightarrow H = \frac{1}{2} a \operatorname{cosec} A \tan \alpha$$

REASONING

51. (b) As, Frugal and Extravagant are antonyms of each other.
Similarly, Criticise and Advocate are antonyms of each other.

52. (a) Pattern of the series is as follows,



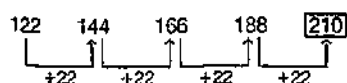


53. (d) As, a can of paint is to be used by paint brush. Similarly, a spool of thread is to be used by sewing needle. Both show the tool needed to perform a task.

54. (d) From Statement I, Manoj > Rahul > Deepak and Vinod From Statement II, Rahul and Manoj > Deepak > Vinod Hence, Statement I alone is sufficient, but Statement II alone is not sufficient.

55. (d) As, on melting, liquid is formed. Similarly on freezing solid is formed.

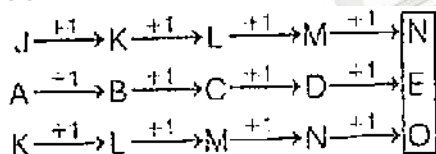
56. (a) Pattern of the series is as follows,



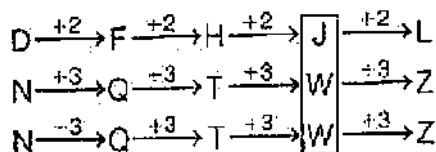
∴ ? = 210.

57. (b) Given, today is Wednesday. Everyday repeats after 7 days, hence Wednesday will repeat after every 7 days. Now, 61 days ÷ 7 = 8 weeks 5 odd days. So, the 5th day after Wednesday is Monday. Hence, Monday is the correct answer.

58. (a) Pattern of the series is as follows.

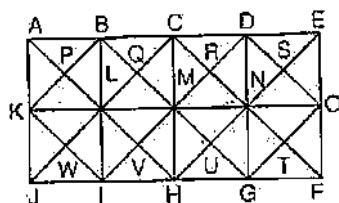


59. (a) Pattern of the series is as follows.



∴ ? = JWW

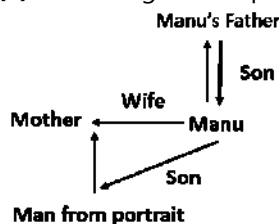
60. (a) Given



Hence, total number of squares = 24 (□ ABKL, □ BCML, □ BQLP, □ KPLW, □ LWIV, □ LQMV, □ QCMR, □ MRNU, □ MVUH, □ RDSN, □ NSOT, □ NTUG, □ BKMI, □ CLNH, □ DMOG, □ ACJH, □ BDGI and □ CEFH)

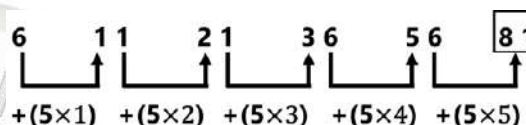
61. (a) 'Catastrophe' is most nearly similar in meaning to the word 'DEBACLE'.

62. (a) According to the question,



Hence, Manu looking at his son's portrait.

63. (c) Pattern of the series is as follows,

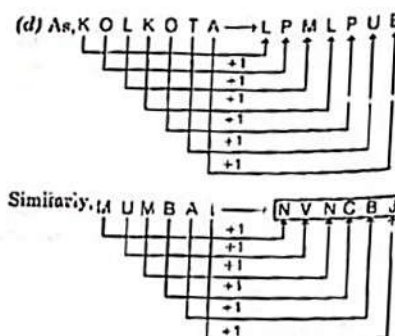


64. (a) according to the question



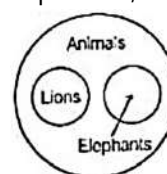
Hence, T is sitting between U and V.

65. (d)



66. (b) As, Butterfly and Freedom are inter-related. Similarly, Self-reliant and Buoyant are inter-related.

67. (a) According to the question,

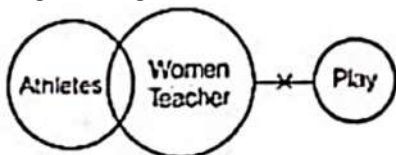


68. (d) 'Alleviate' is opposite in meaning to the given word 'MITIGATE'.

69. (b) Except 'Driving', all other are related to water, while 'Driving' is related to road.

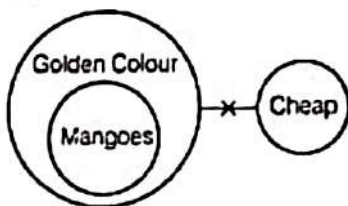


70. (a) According to the given statements,



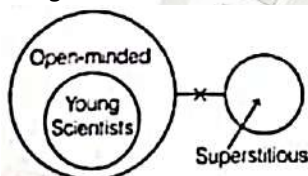
I. (×) II. (✓)
Hence, neither Conclusion I nor II follows.

71. (b) According to the given statements,



I. (×) II. (✓)
Hence, only Conclusion II follows.

72. (b) According to the given statements,



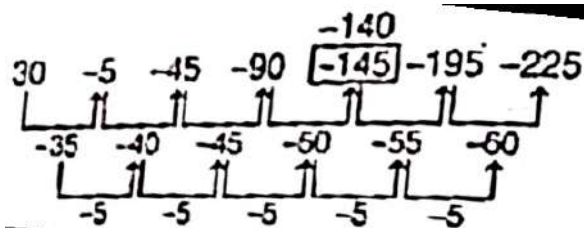
I. (×) II. (×)
Hence, neither Conclusion I nor II follows.

73. (b) According to the question,

Quali tie s/Bo des	Gre en cov er	Yell ow cove r	New volu me	Old volu me	Law repo rts	Medi an extra ct
A	×	✓	✓	×	✓	×
B	✓	×	✓	×	✓	×
C	✓	×	×	✓	✓	×
D	×	✓	✓	×	×	✓
E	✓	×	×	✓	×	✓
F	✓	×	×	✓	×	✓

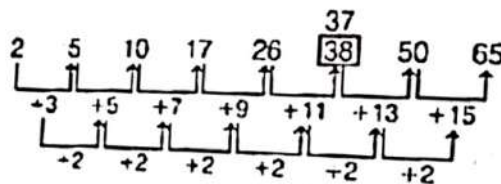
Hence, E and F are two old medical extracts and have covers.

74. (a) Pattern of the series is as follows,



∴ Wrong number = -145

75. (b) Pattern of the series is as follows,



∴ Wrong number = 38

76. (c) Total number of students studying Arts in Institutes and G together 12

$$= 3800 \times \frac{15}{100} + 3800 \times \frac{12}{100}$$

$$= \frac{3800}{100} \times 27$$

$$= 38 \times 27 = 1026$$

77. (c) Total number of students from Institute B study Arts and Commerce

$$= 3800 \times \frac{8}{100} + 4200 \times \frac{17}{100}$$

$$= 38 \times 8 + 42 \times 17$$

$$= 304 + 714 = 1018$$

78. (c) Required ratio

$$= \frac{\text{Number of students studying Commerce in Institute}}{\text{Number of students studying Arts in Institute E}}$$

$$= \frac{3800 \times \frac{14}{100}}{4200 \times \frac{18}{100}}$$

$$= \frac{38 \times 14}{42 \times 18} = \frac{19}{27} = 19 : 27$$

79. (d) According to the given statement, all of three cause actions are follows.

80. (d) According to the given statements, only I and III follows.

Solutions (Q. Nos. 31-35)

According to the given information

Hours	Dramas
1 st	B
2 nd	D
3 rd	Break
4 th	C
5 th	A
6 th	E

81. (c) 3rd hour is a break hour.

82. (c) Drama B is to be staged first.

83. (d) Drama C is to be staged immediately after the break.

84. (a) Break hour happens immediately after D.

85. (d) Drama E is to be staged last.

86. (a) Let age of mother be x yr.



Then, age of person = $\frac{2x}{5}$

According to the question, after 8 yr

$$\frac{1}{2} \times (x + 8) = \frac{2x}{5} + 8$$

$$x + 8 = 2 \left(\frac{(2x+40)}{5} \right)$$

$$5x + 40 = 4x + 80$$

$$= 40 \text{ yr}$$

Hence, present age of Mother 40 yr

87. (d) From, $\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$

Here, $M_1, M_2 = 10, H_1 = 1, H_2 = 4$

$$W_1 = 270 \text{ and } W_2 = ?$$

Now, $\frac{6 \times 1}{270} = \frac{10 \times 4}{W_2}$

$$W_2 = \frac{40 \times 270}{6}$$

$$W_2 = 1800$$

88. (d) Here, $n = 3$ and $(n + 1) = 4$

According to the formula,

The two hands will coincide at $\frac{60n}{11}$ min past n or $\frac{60 \times 3}{11}$

min past 3 or $16 \frac{4}{11}$ min past 3.

89. (a) Required number = HCF of $(1657 - 6) = 1651$ and $(2037 - 5) = 2032$

By division method,

$$1651 \overline{)2032(1}$$

$$\underline{1651}$$

$$381 \overline{)1651(4}$$

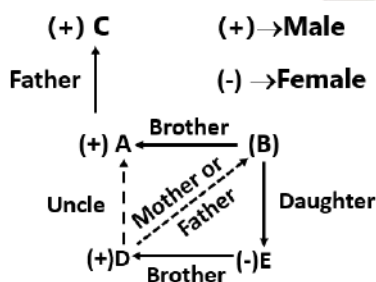
$$\underline{1524}$$

$$127 \overline{)381(3}$$

$$\underline{381}$$

$$\underline{X} = 127$$

90. (d) According to the questions,



Hence, A is (maternal or paternal) uncle of D.

COMPUTER

91. (c) Let $F(A, B, C) = AB + AB' + A'C + AC$

$$F(A, B, C) = A(B + B') + C(A' + A)$$

$$= A(1) + C(1) = A + C$$

Since, the minimised Boolean expression does not contain Boolean variable B . Therefore, $F(A, B, C)$ is unaffected by the value of the Boolean variable B .

92. (d) n bits binary number requires d -decimal digits

So, $10^d > 2^n$

Take log on both side

$$\log_{10} 10^d > \log_{10} 2^n$$

$$d > n \log_{10} 2$$

93. (c) FFFF will be the last memory location in a memory size 64 K.

The equivalent addressing ranges are 0 to 65535 in decimal and 0 to FFFF in hex.

Hexadecimal (also base 16) is a positional numeral system with a radix or base of 16. It uses sixteen distinct symbols, most often the symbols 0 – 9 to represent values zero to nine and A – F to represent values ten to fifteen.

Bytes	Hex	Address range
16	10	0 – F
256	100	0 – FF
1 K	400	0 – 3FF
4 K	1000	0 – FFF
64 K	10000	0 – FFFF

94. (a) $(x + y)(x + y')(x' + y)$
 $= [(x + y)(x + y')](x' + y)$
 $= [x + xy' + xy + yy'](x' + y)$
 $= [x + xy' + xy + 0](x' + y)$
 $= [x + x(y' + y)](x' + y)$
 $= [x + x(1)](x' + y)$
 $= [x + x](x' + y)$
 $= x(x' + y)$
 $= xx' + xy$
 $= 0 + xy = xy$

95. (b) Floating point representation is used to represent real numbers. This representation does not reserve a specific number of bits for the integer part or the fractional part.

The floating number representation of a number has two part : the first part represents a signed fixed point number called Mantissa and the second part of designates the position of the decimal point and is called the exponent.

96. (b) We know that, frequency is defined as the number of cycles in one second.

$$\text{Number of cycle in 1sec} = 2500 \text{ million}$$

$$\text{Frequency} = 2500 \text{ MHz}$$

We know that time period is the inverse of frequency and is defined as the time taken by one cycle.

$$\Rightarrow T = 1/F$$

$$= \frac{1}{2500 \times 10^6}$$

$$= \frac{100}{2500 \times 10^6 \times 100} = \frac{100}{25 \times 10^{10}}$$

$$= \frac{100}{25} \times 10^{-10} = 4 \times 10^{-10} \text{ sec}$$

97. (b) Let $F(A, B, C) = AB + AB'C + AB'C'$
 $= AB + AB'(C + C')$
 $= AB + AB'(1) = AB + AB'$
 $= A(B + B') = A(1) = A$
 So, there is no required of NAND gate.



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98.(c) The maximum and minimum value represented in signed 16 bit 2's complement representations are -32768 and 32767 .

99. (b)

$$\begin{aligned}
 & (A + C) (A + AD) + AC + C \\
 &= (A + C) \cdot A(1 + D) + AC + C \\
 &= (A + C) \cdot A + AC + C \\
 &= A \cdot A + AC + AC + C \\
 &= A + AC + AC + C \\
 &= A + AC + C = A(1 + C) + C = A + C
 \end{aligned}$$

100. (c) In NAND gate, output is high if any of the input is low. The truth table for the NAND gate is:

ENGLISH

101.	c	102.	b	103.	b	104.	d	105.	c
106.	b	107.	c	108.	b	109.	a	110.	b
111.	b	112.	a	113.	c	114.	c	115.	b
116.	d	117.	b	118.	a	119.	a	120.	a

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